

Design of a Slot-Engineered Compact Wideband Self-Isolated Multiple Input Multiple Output Antenna for 5G Millimeter-Wave Applications

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Abstract. We introduce a compact, self-isolated MIMO antenna tailored for 5G millimeter-wave use that relies on slot engineering of the radiating patch. Why add bulky circuitry? A few simple cuts — U-shaped, H-shaped and circular slots — widen the impedance bandwidth and tame surface-current coupling, all without external decoupling elements. Fabricated on an ultra-low-loss substrate and tuned in CST Studio Suite, the design achieves a 4.59 GHz bandwidth centered near 32 GHz. In a 1×2 MIMO layout it provides isolation better than -25 dB (-29 dB at resonance), a peak gain of 8.65–8.85 dBi, 86.8% radiation efficiency, an extremely low ECC 0.00004, and TARC below -18 dB. Expanding to 2×2 boosts isolation further (better than -45 dB), yields a peak gain of 9.36 dBi, 87.89% radiation efficiency, a worst-case ECC 0.00018, and TARC under -18.5 dB. With stable patterns, strong diversity figures and good impedance matching, this small footprint solution is a practical, space-saving option for 5G mmWave MIMO.

Keywords: 5G mm wave· Multiple Input Multiple Output antenna· self-isolated antenna· mutual coupling reduction· slot-based antenna design

1 Introduction

The need for faster data transfer rates, lower latency, and better RF performance across all frequencies is being driven by developments in fifth generation (5G) wireless communication technologies. Because they have a lot of bandwidth available, you will find a variety of new frequency bands in the millimeter-wave (mm-wave) range that are suitable to meet these demands. One drawback of employing numerous antennas for small end-user devices is that each antenna element is positioned quite close to the others, which can make it difficult for them to cooperate effectively due to their mutual coupling. High mutual coupling between neighboring antenna elements will lower each element's performance levels, which will negatively impact the antenna's overall performance.

The literature currently in publication employs a variety of methods to improve isolation in order to overcome these difficulties. As demonstrated in [16]–[20], some conventional methods to lessen the coupling between antennas include the use of electromagnetic bandgap (EBG) structures, defective ground structures (DGS), neutralization lines, and metamaterials. Although these methods work well to increase isolation between antennas, they usually require more material and substrate area than a compact system, which makes it more difficult to produce a small one due to production cost hurdles.

Self-isolating and self-decoupling Multiple Input Multiple Output antenna designs have gained popularity recently as a different strategy for attaining high performance in the Multiple Input Multiple Output arena. There is no need for an external decoupling component because the self-isolation method integrates the isolation mechanism into the antenna’s radiating structure. A variety of methods, such as radiator sharing, current cancellation, folding sub-structure cavities, and hybrid tuning structures, have been used to build self-isolated Multiple Input Multiple Output antenna designs [1–5, 8, 14, 15]. By controlling the distribution of surface currents and removing the coupling between the antennas near each other, systems with slot-type geometries have shown remarkable performance at mm-wave frequencies [3, 6, 9, 12, 13].

According to recent research, Multiple Input Multiple Output antennas for mmwave frequencies and 5G networks feature high levels of inter-element isolation (IEI) and a broad bandwidth. Compact self-isolating slot antennas with low ECC and notable IEI gains have also been reported [3, 4]. Similarly, a number of methods looking at shared aperture and multi-port setups are appropriate for or produced for usage in mobile and portable phone platforms [5,14,19]. Additionally, good performance in terms of IEI and radiation efficiency has been made possible by the use of SIW technology and cavity designs at both 28GHz and 38GHz [12,25].

We demonstrate a compact, wideband MIMO antenna with inherent isolation that spans the 5G millimeter-wave range from 21 to 45 GHz. Rather than bolting on external decouplers, the design carves intentional slot features directly into the radiating elements; those etched grooves reroute surface currents so the array isolates itself — imagine snipping tiny channels into pavement to steer traffic. No extra boxes, no fiddly retuning. The result is an integrated isolation approach that removes added hardware, yielding a simpler, smaller antenna that fits neatly into 5G mmWave installations.

We tested the antenna on the bench in three setups — single element, MIMO 1×2 and MIMO 2×2 — measuring S-parameters, isolation, radiation patterns, the Envelope Correlation Coefficient (ECC) and the Total Active Reflection Coefficient (TARC). The data show a wide impedance bandwidth and consistently high isolation. Radiation patterns remain stable; ECC and TARC stay low. Those are textbook signs of strong diversity performance and good impedance matching — which, as any RF engineer will tell you, is harder to pull off than it sounds — even when multiple ports are driven at once. These traits make the design a compelling candidate for 5G millimeter-wave systems.

2 Antenna Design and Methodology Suggestions

This section covers the geometry, isolation techniques, and design process of the new self-isolated Multiple Input Multiple Output antenna developed for 5G millimeter waves, achieving an extensive bandwidth and providing substantial separation between closely spaced elements while maintaining a small, planar design format.

2.1 Configuration of the Antenna

We chose Rogers RT/duroid 5870 as the substrate — its low dielectric loss and dependable millimeter-wave behavior make it a natural fit for high-frequency antennas. The material's permittivity ($\epsilon_r = 2.33$), 0.508 mm thickness (about half a millimeter) and loss tangent of 0.0012 are numbers that help keep the response predictable across the band. The radiator is a rectangular microstrip patch, fed by a microstrip line and backed by a continuous ground plane — a simple, stacked arrangement that keeps the design compact and straightforward to fabricate. Initial patch dimensions came from standard microstrip formulas and were then fine-tuned, over a few iterations, with full-wave CST Studio Suite simulations to cover 21–45 GHz.

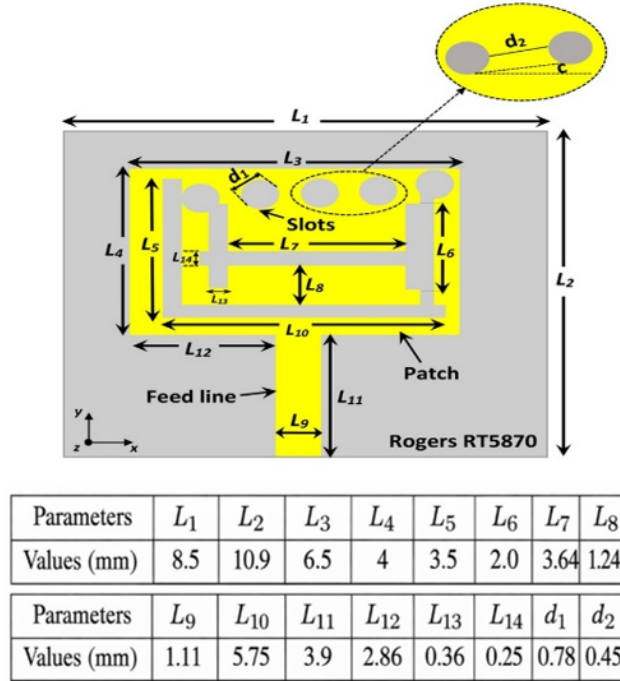


Fig. 1: Dimensions of MIMO Antenna

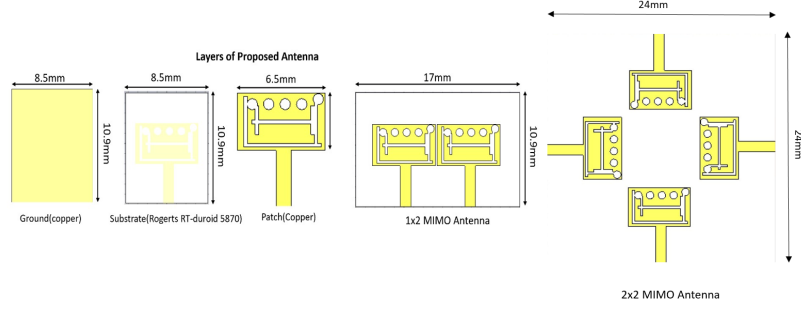


Fig. 2: The suggested single element, 1x2, and 2x2 Multiple Input Multiple Output antennas' geometries and configurations

2.2 Design Evolution Using Slot Engineering

A methodical slot-engineering technique is used to increase the impedance bandwidth and reduce mutual coupling. The reference configuration for the design evolution is a traditional rectangular patch antenna without slots. When expanded to Multiple Input Multiple Output setups, this structure shows high surface current interaction and restricted bandwidth, although exhibiting resonance within the desired frequency range.

To disrupt the predominant surface current routes and add a second resonant mode, a U shaped slot is placed on the radiation performance patch during the initial modification stage. By using this design, the impedance bandwidth is increased and the current concentration at the patch's edges is decreased. The surface current flow will then be further disrupted by the installation of H-shaped slots, which will also spread the surface current across the radiator. Finally, the creation of multiple circular slots in areas that exhibit high levels of surface current concentration will further create discontinuities in the surface current path, increasing the overall bandwidth and isolation performance of the antenna.

2.3 Self-Isolation Mechanism

By purposefully changing the distribution of surface currents on the antenna through the insertion of slot structures, this antenna achieves self-isolation. Closely spaced elements are mutually connected due to surface and near-field coupling effects from element placement; this design's slot lowers the amount of energy coupled between elements. which serve as current flow blocking and redirection components.

By positioning the slots directly in the radiating element, this design offers an integrated solution in addition to the traditional methods of achieving isolation, which depend on adding an external structure, such as electromagnetic

bandgap surfaces, metasurfaces, and defective ground planes. We can produce considerably more compact antennas with fewer design steps and, as a result, lower manufacturing overhead by using this kind of integrated method.

2.4 Multiple Input Multiple Output Configurations

In order to show that this design would be a feasible choice for tiny millimeter wave applications, the goal of this study is to assess how well the planned Multiple Input Multiple Output antenna performs when equipped with 1x2 Multiple Input Multiple Output and 2x2 Multiple Input Multiple Output. As would be common in a real-world 5G millimeter-wave deployment, the antenna's design relied on close spacing between neighboring pieces.

To get insight into the antennas' performance in terms of impedance characteristics, radiation pattern stability, and isolation characteristics over the whole operating frequency range, the designs were simulated in CST Studio Suite.

Simulation results verified that the slot-based design provides strong isolation across feeds in both 1x2 and 2x2 configurations while still achieving broadband performance. As a result, the slot-based antenna is also thought to be suitable for small millimeter-wave Multiple Input Multiple Output applications.

3 Simulation Results and Discussion

The performance of the Multiple Input Multiple Output antenna design in 1x2 and 2x2 configurations was simulated using CST Studio Suite. These findings were examined in terms of radiation, isolation, impedance match, and overall performance.

3.1 Impedance Bandwidth and S-Parameters

The S-parameters for the simulated antenna displayed here demonstrate the antenna's dependability since the S11 (reflection coefficient) stays below -10 dB, showing good matching between the antenna and its feed line/power source across a broad frequency range. Furthermore, a low Transmission Coefficient (S21) shows that the two antenna elements are still sufficiently isolated and do not interact with one another within the primary operational frequency range.

Figure 3 displays the 2 × 2 Multiple Input Multiple Output antenna's simulated S-parameters. The antenna exhibits better isolation and good impedance matching when compared to the 1 × 2 design. The observation of many resonant modes facilitates wideband operation in the millimeter-wave frequency range.

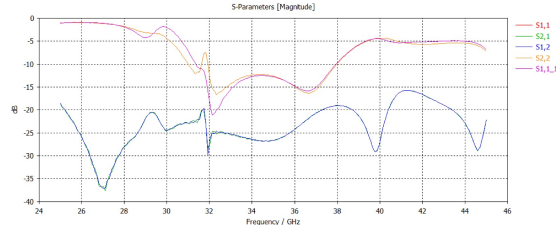


Fig. 3: Simulated S-parameters of the proposed 1×2 Multiple Input Multiple Output antenna

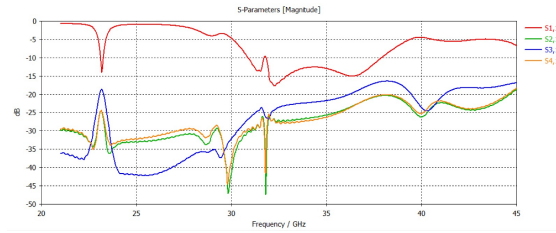


Fig. 4: Simulated S-parameters of the proposed 2×2 Multiple Input Multiple Output antenna

3.2 Isolation Performance

The isolation performance of the proposed antenna is evaluated using the mutual coupling characteristics. At 32 GHz, the 1×2 Multiple Input Multiple Output design achieves an isolation level of roughly -25 dB. With a peak isolation of roughly -45 dB at the same frequency, the 2×2 Multiple Input Multiple Output setup shows a notable improvement. This enhancement shows how well the slot-engineered radiator suppresses surface current coupling between neighboring antenna elements.

Multiple Input Multiple Output Configuration	Frequency (GHz)	Isolation (dB)
1×2 Multiple Input Multiple Output	32	-25
2×2 Multiple Input Multiple Output	32	-45

Table 1: The suggested Multiple Input Multiple Output antenna's isolation performance

3.3 Radiation Performance

Fig. 5 displays the calculated 3D radiation patterns of the proposed 1×2,2×2 Multiple Input Multiple Output antennas at representative frequencies of 32 and 33Ghz Respectively. The antenna confirms dependable radiation performance across wideband millimeter-wave operation by maintaining consistent broadside radiation throughout the operational band with no pattern distortion.

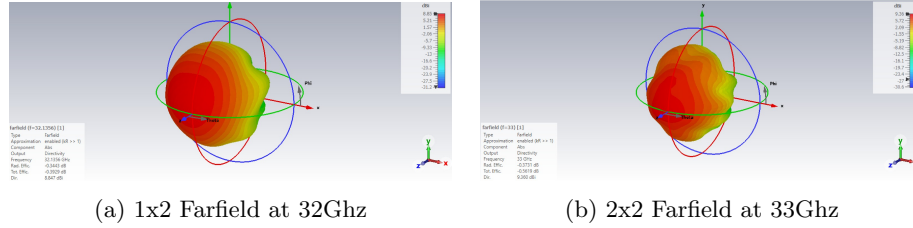


Fig. 5: The suggested Multiple Input Multiple Output antenna's simulated 3D far-field radiation patterns at various operating frequencies

3.4 Performance Comparison

Table 1 compares the performance characteristics of the proposed 1×2 and 2×2 Multiple Input Multiple Output antennas. In conclusion, compared to the 1×2 setup, the 2×2 Multiple Input Multiple Output antenna has a greater bandwidth, better gain, and less isolation. Furthermore, both configurations demonstrated outstanding radiation efficiencies, demonstrating that the suggested slot-based self-isolation technique boosted Multiple Input Multiple Output performance while preserving favorable radiation characteristics.

Parameter	1×2 MIMO Antenna	2×2 MIMO Antenna
Bandwidth (GHz)	4.4594	5.479
Isolation (dB)	-25 at 32 GHz	-45 at 32 GHz
Gain (dBi)	8.65–8.85	9.36
Radiation Efficiency (%)	86.8	87.89

Table 2: Comparing the suggested 1×2 and 2×2 Multiple Input Multiple Output antennas' performance

4 Multiple Input Multiple Output Analysis

We assess the MIMO performance of the proposed antenna using two practical metrics: the Envelope Correlation Coefficient (ECC) and the Total Active Reflection Coefficient (TARC). These two measures capture complementary aspects

of multiport behavior — ECC describes diversity potential, while TARC reveals how the antenna’s impedance behaves when more than one port is driven. Both parameters were calculated from the simulated S-parameters exported from CST Studio Suite for the 1×2 and 2×2 MIMO layouts.

4.1 Envelope Correlation Coefficient (ECC)

The Envelope Correlation Coefficient (ECC) quantifies how similar the radiation patterns of different antenna elements are. A low ECC means the elements act independently and the system can exploit diversity effectively. Put simply: lower is better. Here ECC is computed from S-parameters using the standard formulation [8]: [

$$\text{ECC} = \frac{|S_{11}S_{12}^* + S_{21}S_{22}^*|^2}{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)} \quad (1)$$

] For the proposed 1×2 MIMO antenna the ECC is on the order of 4×10^{-5} at the operating frequency near 32 GHz. In the 2×2 configuration the worst-case ECC across the band is about 1.8×10^{-4} . To put those numbers in perspective, they sit orders of magnitude below the commonly used practical threshold of 0.01 (and are far below the theoretical limit of 0.5). In practical terms, such ultra-low ECC indicates almost negligible correlation between elements — the slot-engineered, self-isolated radiators suppress surface-current coupling so effectively that each element behaves almost as if it were radiating on its own. Think of it as well-behaved roommates who rarely borrow each other’s stuff.

4.2 Total Active Reflection Coefficient (TARC)

TARC captures the effective impedance match of a MIMO antenna when all ports are driven simultaneously with equal amplitudes but possibly different phases. Unlike single-port return loss, TARC accounts for both reflections and mutual coupling, giving a more realistic picture of matching under active multi-port excitation. The metric is computed as [8]: [

$$\text{TARC} = \sqrt{\frac{|S_{11} + S_{12}e^{j\theta}|^2 + |S_{21} + S_{22}e^{j\theta}|^2}{2}}, \quad (2)$$

] where θ denotes the phase difference between the excitation signals at the ports. Simulation results show that for the 1×2 MIMO antenna the TARC stays below -18 dB across the operating band; the 2×2 layout performs slightly better, with TARC values better than -18.5 dB. Those low TARC levels imply the antenna maintains excellent impedance matching and efficient power transfer even when multiple ports are active. In other words — despite simultaneous excitation, the active input impedances remain stable, a direct consequence of the slot engineering that minimizes mutual coupling. Overall, the combination of ultra-low ECC and consistently low TARC demonstrates that the proposed antenna offers strong diversity, high isolation, and robust impedance matching — traits that make it well suited for high-capacity 5G millimeter-wave MIMO deployments.

5 Conclusion

The creation of a small, weatherproof multi-input/multi-output antenna for use in Fifth Generation (5G) millimeter wave frequency bands is covered in this study. Rogers RT/Duroid 5870 substrate material, which has a useful frequency range of 21 to 45 GHz, is used in the construction of the antenna. Its innovative use of slot-engineered radiation patches and U-, H-, and circular slot designs allows for accurate impedance tuning and improved isolation between multiple antennas positioned close together, eliminating the need for external decoupling structures.

Simulation has shown that this antenna performs exceptionally well in terms of isolation. The antenna's consistent radiation patterns in both Multiple Input Multiple Output configurations (1x2 and 2x2) are most notable, though. This antenna offers dependable communication because of its high reflection coefficients and minimal envelope correlation. For upcoming fifth-generation (5G) millimeter waves, the antenna will be a great choice.

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